RES POLYTECHNIC, BANGALORE-91

CIE Practice Test II

Program	Computer Science and Engineering		Semester- III
Course Name	Data Structure with Python		Test II
Course Code	25CS33I	Duration- 180 min	Marks-50
Name of the Course Coordinator: Vinay Kumar			
Note: Answer any one full question from each section. Each full question carries equal marks. Write the program, execute it and show the output to the examiner.			

Q. No	Questions	CL	CO	Marks
	SECTION I			
1	**a)** Write a Python program, using functions, to perform the following operations on a dictionary of students (name as key, marks as value): **(12 marks)** 1. Add a new student and update the marks of an existing student. 2. Delete a student entered by the user (display a message if the student is not present). 3. Display the keys using keys(), the marks using values() and all the entries using items(). 4. Iterate through the dictionary and display each name with its marks. **Example** Input: {'Ravi': 72, 'Sita': 85} Add: 'Arun': 60 Output: {'Ravi': 72, 'Sita': 85, 'Arun': 60} **b)** Write a Python program to perform the following file operations: **(13 marks)** 1. Create a file named students.txt and write 5 student names into it. 2. Read and display the contents of the file. 3. Append two more names and display the file again. 4. Display the file attributes (name, mode and whether the file is closed).	L3	1, 2, 3	25
	OR			
2	**a)** Write a Python program to perform the following: **(12 marks)** 1. Create two sets and display their union, intersection, difference and symmetric difference. 2. Create a tuple, access its elements using indexing and slicing, and display the result. 3. Pack the values name, age and city into a tuple and unpack them into separate variables. **Example** set1 = {10, 20, 30, 40}, set2 = {30, 40, 50} Union: {10, 20, 30, 40, 50} Intersection: {30, 40} **b)** Write a Python program to perform the following: **(13 marks)** 1. Create a list of 6 numbers and display it using indexing and slicing. 2. Append a number, remove a number entered by the user, and sort the list. 3. Using a list comprehension, create a new list containing the squares of the even numbers. 4. Create a string, display its characters using indexing and slicing, and display the result of concatenation, repetition and any three built-in string methods.	L3	1, 2, 3	25

Q. No	Questions	CL	CO	Marks
	SECTION II			
3	**a)** Write a Python program that reads two numbers from the user and displays their division. Use exception handling to: **(12 marks)** 1. Handle ZeroDivisionError and ValueError with suitable messages. 2. Use the else block to display the result and the finally block to display "Execution completed". Display one example each of a syntax error, a runtime error and a logical error. **Example** Input: 10 and 0 Output: Cannot divide by zero **b)** Write Python programs to find the factorial of a number: **(13 marks)** 1. Using a recursive function, clearly showing the base case and the recursive case. 2. Using iteration (loop). Display the output of both programs for the same number to show the difference between recursion and iteration. **Example** Input: 5 Output: 120	L3	1, 2, 3	25
	OR			
4	**a)** Write a Python program to create a User-Defined Data Type (class) **Student** with the following: **(12 marks)** 1. A constructor to store the name, roll number and marks. 2. A method to display the student details. 3. A method to update the marks. Create two objects of the class and show the working of all the methods. **b)** Write a Python program to implement a List ADT using a class (UDT) with the following operations: **(13 marks)** 1. insert(item) to add an item at the end. 2. delete(item) to remove an item (display a message if it is not present). 3. search(item) to check whether an item is present. 4. display() to show all the items.	L3	1, 2, 3	25

Note for the Course Coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.

Sign of the Course Coordinator Signature of the HOD Signature of the IQAC Chairman